



Seminar: Mathematics in Lean

LECTURERS: PROF. DR. JUDITH LUDWIG AND DR. JUNYAN XU
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Preliminary Meeting: Tuesday, February 3, 2:15pm, room SR4.

Time and Place: Tuesdays 2-4pm, room TBD.

Lean is an interactive theorem prover enabling mechanical verification of mathematical proofs with absolute certainty. It combines the programming language of Lean (built on dependent type theory) with an interactive user interface and powerful and extensible automation (tactics). Lean has a mathematical library (mathlib), which is steadily growing as an open-source project with now hundreds of contributors and over 2 million lines of code. This has enabled large-scale formalization projects, such as the Liquid-Tensor-Experiment¹, the Carleson Project or the ongoing project on formalizing Fermat's Last Theorem. The goal of each of these projects was/is to verify mathematical results using a computer, and in particular the proof assistant Lean. For this one has to express all definitions, statements and proofs involved in the theorem prover in an error-free way, a process known as formalization of mathematics.

In this seminar you'll learn how to formalize mathematics in Lean by working through the tutorial Mathematics in Lean. It gives an introduction to formalizing mathematics through a learning by doing approach to Lean. It also explains the general design of mathlib and introduces mathematics already formalized there. Each talk will consist of presenting the content of one of the Chapters 2-11 and explaining some of the exercises in a Lean file. In addition to your talk, you will work on a small formalization project. By completing the seminar you will acquire the skills needed to join a bigger formalization project or to contribute to mathlib.

¹See also this article.

Practicalities

1. **Prerequisites:** You should have a good understanding of basic algebra, analysis and topology.

2. **Preparations for participants:**

- Play the Natural Number Game on the Lean Game Server.
- Familiarize yourself with basic Git operations, including creating branches, committing changes, pushing to a remote repository, and opening pull requests.
- The seminar repository is located on GitHub. Please make sure you get access; how to do this will be explained in the preliminary meeting.
- Please also register an account on the Lean Zulip chat. If you require technical assistance, you can send a direct message to Junyan Xu on Zulip.

3. **Requirements for participants:**

- **Talk:** You are expected to give a 60-minute talk on your chosen chapter of *Mathematics in Lean*. Students' talks start from Chapter 2 in the second week; the first week's meeting on April 14th will feature an introductory talk on Lean, mathlib, and dependent type theory. During the last 30 minutes of each session participants will work on the exercises of the respective chapter together. As an expert on that chapter, you should help answer questions that arise.
- Before the talk, please solve all exercises in the chapter you are going to present and submit your solutions in a branch in the course repository on GitHub or by sharing the Lean file on Zulip or via email. Submitting via the repository will make it easier to get direct feedback. Solutions to some of the exercises should be presented as part of your talk.
- Start preparing your presentation in good time (about 4 weeks in advance). Some general guidance on how to give a good seminar talk can be found here. Two weeks before your presentation, meet Dr. Xu for a preliminary discussion. This helps in identifying gaps in understanding.
- Attendance of the talks is mandatory.
- Language: Your talk must be in English.

Resources

Mathematics in Lean: https://leanprover-community.github.io/mathematics_in_lean/

Official Lean website: <https://lean-lang.org/>

Lean Game Server: <https://adam.math.hhu.de/>

Mathlib overview: <https://leanprover-community.github.io/mathlib-overview.html>

Zulip chat: <https://leanprover.zulipchat.com/>